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Fluorescence *in situ* Hybridization (FISH) analysis of post-treatment vaginal microbial colonization in antibiotic-responsive and antibiotic-resistant bacterial vaginosis

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Project Proposal

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1. Introduction

Bacterial vaginosis (BV) is the most common type of vaginal dysbiosis among women of reproductive age, characterized by a shift from a *Lactobacillus*-dominated microbiota to a diverse community of facultative and obligate anaerobic bacteria (Swidsinski et al., 2024). Key taxa associated with BV include *Gardnerella vaginalis*, *Fannyhessea*, *Prevotella*, *Mobiluncus*, and *Megasphaera* species, along with a significant reduction in lactic acid-producing lactobacilli (Chen et al., 2024). This disturbance leads to an increased vaginal pH and the loss of protective antimicrobial functions usually mediated by lactic acid and hydrogen peroxide production. Clinically, BV presents with symptoms such as abnormal vaginal discharge, odor, and irritation, although many cases are asymptomatic (Scapaticci et al., 2020).

BV is highly prevalent globally, with disproportionately high burdens reported in sub-Saharan Africa (Milford et al., 2026). Importantly, BV is associated with significant reproductive and sexual health complications, including increased susceptibility to HIV acquisition, other sexually transmitted infections (STIs), pelvic inflammatory disease, and adverse pregnancy outcomes such as preterm birth (George et al., 2025). Established risk factors include multiple sexual partners, inconsistent condom use, vaginal douching, smoking, and hormonal fluctuations (Danjuma et al., 2025).

Current standard-of-care treatments rely primarily on antibiotics such as metronidazole and clindamycin (Feuillolay et al., 2025). While these therapies are often effective in the short term, recurrence rates remain unacceptably high, with over 50% of women experiencing relapse within 6–12 months (Delfstra et al., 2023). This high recurrence is largely attributed to the persistence of BV-associated polymicrobial biofilms, particularly those structured by *Gardnerella* spp., which exhibit increased tolerance to antimicrobial agents compared to planktonic bacteria (Munch et al., 2024). The inability of current treatments to fully restore a stable, *Lactobacillus*-dominated microbiome further underscores the need for improved therapeutic strategies (Li et al., 2024).

In response, there is growing interest in alternative and adjunctive BV treatments, including probiotics, live biotherapeutics, microbiome-modulating interventions, and anti-biofilm agents (Brotman et al., 2014, O’Hanlon et al., 2013). Studies investigating these emerging treatment strategies primarily rely on standard molecular approaches, such as qPCR and 16S rRNA sequencing, to characterize vaginal microbial communities (Rosca et al., 2020). However, these

techniques do not preserve the spatial context required to investigate microbial organization and host-microbe interactions (Swidsinski et al., 2005).

Fluorescence in situ hybridization (FISH) is a powerful yet underutilized tool in this field, enabling the identification and localization of specific microorganisms within intact samples through the hybridization of fluorescently labeled probes to ribosomal RNA (Savicheva, 2023). Unlike sequencing-based approaches, FISH allows for the direct visualization of microbial spatial organization, making it particularly valuable for studying polymicrobial biofilms and microbial community structure *in situ* (Machado et al., 2013). Multiplex FISH further expands this capability by enabling the simultaneous detection of multiple bacterial taxa within a single sample, providing a more comprehensive view of microbial composition and spatial relationships. Given the increasing complexity of microbial interactions implicated in BV pathogenesis and treatment response, optimizing multiplex FISH approaches is essential. We hypothesize that antibiotic-resistant BV is associated with distinct post-treatment vaginal microbial colonization and spatial organization, characterized by persistent BV-associated anaerobes and reduced *Lactobacillus* dominance. This study, therefore, aims to optimize and apply multiplex FISH to characterize spatial vaginal microbial communities following treatment, to improve our understanding of post-treatment colonization patterns and inform the development and evaluation of novel BV therapeutics.

2. Literature review

2.1 BV: Current Understanding and Clinical Significance

BV is the most prevalent vaginal dysbiosis affecting women of reproductive age and remains a major public health concern worldwide (Coudray and Madhivanan, 2020). Traditionally, BV was regarded as an infection caused primarily by *Gardnerella vaginalis*; however, advances in molecular microbiology have challenged this reductionist perspective by demonstrating that BV is a polymicrobial condition characterized by depletion of protective *Lactobacillus* species and overgrowth of diverse anaerobic bacteria, including *Prevotella* species, *Mobiluncus* species, *Atopobium vaginae*, and *Mycoplasma hominis* (Thulkar et al., 2012). This ecological disruption is associated with elevated vaginal pH, positive amine test, clue cells, and abnormal vaginal discharge, which collectively form the basis of the Amsel diagnostic criteria (Armstrong and Wilson, 2010). Despite decades of research, the initiating events leading to BV remain incompletely understood, generating debate regarding whether BV represents a sexually

transmitted condition, a disturbance of host–microbial homeostasis, or a biofilm-mediated polymicrobial infection (Bilardi et al., 2013, Surapaneni et al., 2021). Increasing evidence now supports the argument that BV is not caused by a single microorganism but rather by cooperative interactions among multiple bacterial species that establish persistent dysbiotic biofilms (Cerca, 2019, Castro et al., 2022b).

The epidemiology of BV further highlights the complexity of the condition. Prevalence differs considerably across populations and geographical regions, with women in sub-Saharan Africa exhibiting some of the highest global rates (Bautista et al., 2016). Additionally, NHANES data showed that approximately 29.2% of women of reproductive age in the United States were BV-positive, yet most cases were asymptomatic (Riaz et al., 2025). This asymptomatic presentation has raised important clinical and scientific debates regarding disease progression, persistence, and treatment necessity. Some studies suggest asymptomatic BV may represent a transient microbial imbalance, whereas others argue that untreated asymptomatic BV contributes to chronic dysbiosis and reproductive complications (Bilardi et al., 2013). Importantly, BV has consistently been associated with pelvic inflammatory disease, infertility, preterm birth, and increased susceptibility to sexually transmitted infections such as Human Immunodeficiency Virus (HIV), gonorrhea, trichomoniasis, and Herpes Simplex Virus Type 2 (Forcey et al., 2015, Bitew et al., 2017). These associations suggest that BV is not simply a superficial alteration in microbial composition, but rather a clinically significant dysbiosis with profound reproductive consequences (Abbe, 2023). However, while previous research has largely focused on identifying microbial taxa associated with BV, relatively limited attention has been directed toward understanding how BV-associated microorganisms spatially colonize and reorganize after antibiotic treatment. This gap is particularly important because persistent microbial organization may contribute directly to treatment failure and recurrence (Bradshaw and Brotman, 2015)

2.2 Vaginal Microbiota in Health and Dysbiosis

The healthy vaginal microbiota is generally dominated by *Lactobacillus species*, particularly *Lactobacillus crispatus*, *Lactobacillus gasseri*, *Lactobacillus iners*, and *Lactobacillus jensenii*, which maintain vaginal homeostasis through production of lactic acid, hydrogen peroxide, bacteriocins, and biosurfactants (Chee et al., 2020). Using 16S rRNA sequencing, researchers classified vaginal microbiota into five major community state types (CSTs), with CST I, II, III,

and V dominated by lactobacilli, whereas CST IV is characterized by reduced lactobacilli and enrichment of strict anaerobes associated with BV (Nunn and Forney, 2016). Although sequencing technologies have significantly improved understanding of vaginal microbial diversity, debate persists regarding the role of specific *Lactobacillus* species in protecting against BV recurrence. For example, *Lactobacillus crispatus* is generally associated with microbial stability and reduced recurrence, whereas *Lactobacillus iners* is often linked to transitional microbial states and increased susceptibility to dysbiosis ((Nunn and Forney, 2016, Forcey et al., 2015). This suggests that vaginal health is determined not only by bacterial presence but also by microbial function, ecological interactions, and colonization dynamics.

Current evidence indicates that the transition from health to BV is not merely a replacement of lactobacilli with anaerobes but rather a complex ecological restructuring involving biofilm development and interspecies cooperation (Chen et al., 2022). Studies have shown that virulent strains of *Gardnerella vaginalis* adhere to vaginal epithelial cells and facilitate colonization by secondary anaerobes such as *Prevotella bivia* and *Atopobium vaginae* (Onderdonk et al., 2016). Castro et al. (2022) further demonstrated metabolic cooperation between *Gardnerella* and *Prevotella* species that enhances biofilm maturation and persistence. Similarly, Gilbert et al. (2019) identified enrichment of *Gardnerella* and *Prevotella* species in recurrent BV, suggesting that microbial interactions play a central role in post-treatment persistence. These findings have shifted scientific focus from individual pathogens toward understanding polymicrobial communities and their structural organization within the vaginal environment. Nevertheless, most current studies rely heavily on sequencing technologies that characterize microbial abundance but provide limited insight into the spatial localization and microbial architecture associated with persistence and recurrence. Consequently, important questions remain regarding how microbial communities reorganize following antibiotic exposure and whether recurrent BV is associated with specific spatial colonization patterns.

2.3 Antibiotic Treatment Failure and Microbial Persistence in BV

Metronidazole and clindamycin remain the recommended first-line therapies for BV because of their broad-spectrum activity against anaerobic bacteria (Vodstrcil et al., 2021). Short-term cure rates of approximately 80–90% have been reported following treatment; however, recurrence remains one of the greatest challenges in BV management (Gao et al., 2024). Studies

indicate that up to 60–70% of women experience recurrence within 3-12 months after treatment, suggesting that current therapeutic approaches fail to restore long-term microbial stability (Thulkar et al., 2012). This discrepancy between short-term symptom resolution and long-term recurrence has generated considerable debate regarding the mechanisms underlying treatment failure. One perspective argues that antibiotics effectively eliminate planktonic bacteria but fail to eradicate adherent polymicrobial biofilms embedded within the vaginal mucosa (Muzny and Sobel, 2023). Supporting this argument, Swidsinski et al. (2008) demonstrated persistence of dense adherent biofilms following metronidazole treatment, indicating that bacterial reservoirs may survive therapy and contribute to recurrent infection (George et al., 2026).

Another major area of debate concerns the role of antimicrobial resistance and bacterial adaptation in recurrent BV. Paunkov et al. (2025) demonstrated that reduced expression of the GvNR1 nitroreductase enzyme contributes to metronidazole resistance in *Gardnerella vaginalis*. Similarly, Castellano et al. (2025) reported increased abundance of antimicrobial resistance genes among diverse BV-associated microbial communities, particularly within *Gardnerella* and *Prevotella* species. Li et al. (2024) further proposed that biofilm-associated resistance involves multiple mechanisms, including restricted antibiotic penetration, extracellular polymeric matrices, altered metabolic activity, efflux pumps, and genetic adaptation. Collectively, these findings suggest that recurrent BV is driven not only by microbial composition but also by the structural organization and persistence of bacterial biofilms following treatment. However, most previous studies investigating recurrence have focused primarily on microbial abundance and resistance genes, with limited exploration of how microbial spatial organization changes after antibiotic exposure. This limitation represents a significant research gap because persistent bacterial architecture may influence antibiotic tolerance, recolonization, and long-term recurrence (Paunkov et al., 2025, Castellano et al., 2025).

2.4 Molecular Approaches Used in BV Research

Molecular approaches such as quantitative polymerase chain reaction (qPCR) and 16S rRNA gene sequencing have significantly advanced understanding of vaginal microbial composition and diversity in BV. These techniques enable sensitive detection and quantification of BV-associated bacteria and have contributed substantially to the identification of microbial

signatures associated with health, dysbiosis, treatment response, and recurrence (Muzny et al., 2023, Castro et al., 2022a). In particular, 16S rRNA sequencing has facilitated characterization of complex polymicrobial communities that cannot be adequately studied using conventional culture-based techniques.

Despite these advances, important limitations remain. Both qPCR and sequencing-based methods require nucleic acid extraction, which disrupts the native spatial arrangement of microorganisms within vaginal samples. As a result, these approaches cannot determine whether bacteria are adherent to epithelial surfaces, dispersed within mucus, or spatially interacting with neighboring microbial species (Muzny and Schwebke, 2016). This limitation is particularly important because microbial localization and structural organization are increasingly recognized as key determinants of persistence, antibiotic tolerance, and recurrence. Consequently, there is a growing need for complementary approaches capable of preserving microbial architecture while simultaneously identifying specific bacterial taxa within intact vaginal communities.

2.5. Fluorescence *in situ* Hybridization (FISH) in BV Research

FISH has emerged as an important cultivation-independent molecular technique for investigating microbial organization within clinical samples (Moter and Göbel, 2000, Lukumbuzya et al., 2019). FISH utilizes fluorescently labelled oligonucleotide probes that hybridize to complementary ribosomal RNA sequences within intact microbial cells, thereby enabling direct visualization and identification of microorganisms while preserving their spatial arrangement (Amann et al., 1995). Unlike culture-based techniques, which are unable to cultivate many fastidious anaerobic bacteria associated with BV, FISH allows microorganisms to be examined directly within intact vaginal biofilms (Amann and Fuchs, 2008). Furthermore, unlike sequencing technologies and qPCR, which require nucleic acid extraction and therefore disrupt microbial architecture, FISH preserves the physical relationships among microorganisms and host epithelial surfaces (Muzny and Schwebke, 2016). This distinction is particularly important because microbial localization and biofilm structure are increasingly recognized as key determinants of persistence, antibiotic tolerance, and recurrence (Lebeaux et al., 2014).

Several comparative studies have highlighted the complementary role of FISH in BV research. Swidsinski et al. (2005) used FISH to demonstrate that BV is characterized by dense

polymicrobial biofilms dominated by *Gardnerella vaginalis* attached to vaginal epithelial cells, thereby fundamentally shifting the understanding of BV from a planktonic infection model to a structured biofilm-associated disease. Subsequent multiplex FISH studies identified co-localization of *Gardnerella vaginalis*, *Atopobium vaginae*, *Prevotella* species, and *Mobiluncus* species within adherent vaginal biofilms, demonstrating complex interspecies interactions and structural microbial organization (Swidsinski et al., 2024, Hardy et al., 2015). While sequencing approaches provide detailed taxonomic profiles, they cannot determine whether microorganisms are adherent to epithelial surfaces, dispersed within mucus, or interacting directly with neighboring species (Muzny and Schwebke, 2016). Consequently, FISH provides unique spatial information that cannot be obtained through sequencing alone.

Despite these advances, important research gaps remain. Most existing FISH studies have focused on untreated BV or general characterization of vaginal biofilms, whereas relatively few longitudinal investigations have examined post-treatment microbial recolonization patterns in women with recurrent versus non-recurrent BV (Hardy et al., 2015, Castro et al., 2022b). Furthermore, limited research has investigated whether persistent adherent biofilms remain after antibiotic therapy in clinically responsive women who later experience recurrence. The lack of spatially resolved post-treatment data limits current understanding of how microbial persistence contributes to recurrent BV and antibiotic failure (Vodstreil et al., 2021). Therefore, integrating optimized multiplex FISH with complementary molecular approaches such as 16S rRNA sequencing may provide a more comprehensive understanding of both microbial composition and spatial organization during post-treatment recolonization (Swidsinski et al., 2024). This approach is particularly relevant to the current study, which aims to characterize and compare post-treatment vaginal microbial colonization in antibiotic-responsive and recurrent BV using optimized multiplex FISH. This approach is particularly relevant to the current study, which aims to characterize and compare post-treatment vaginal microbial colonization in antibiotic-responsive and antibiotic-resistant BV using optimized multiplex FISH.

2.6 Optimization and Applications of Multiplex FISH in BV Research

Multiplex FISH offers several important advantages in the study of vaginal microbial ecology and recurrent BV. The technique enables simultaneous detection of multiple bacterial species within intact biofilms, preserves microbial spatial architecture, and facilitates direct visualization of host-microbe interactions without the need for cultivation (Moter and Göbel, 2000, Lukumbuzya et al., 2019). Additionally, species-specific probes provide high specificity and enable differentiation between closely related microorganisms within polymicrobial communities (Amann et al., 1990). These characteristics make multiplex FISH particularly valuable for investigating post-treatment microbial persistence and biofilm-associated recurrence. However, the technique also has limitations. FISH sensitivity may be reduced in microorganisms with low ribosomal RNA content, probe penetration may be limited in dense biofilms, and fluorescence background interference may complicate image interpretation (Amann and Fuchs, 2008). Moreover, FISH requires specialized fluorescence microscopy and technical expertise and does not provide the broad functional or genomic information obtainable through next-generation sequencing technologies (Amann and Fuchs, 2008).

To address these limitations, several optimization strategies have been proposed. Improved probe design, enhanced permeabilization protocols, signal amplification systems, peptide nucleic acid (PNA) probes, and integration with confocal laser scanning microscopy have all been shown to improve probe specificity, sensitivity, and biofilm visualization (Stender et al., 2002, Lukumbuzya et al., 2019). Recent studies have also recommended combining FISH with molecular approaches such as 16S rRNA sequencing and qPCR to integrate spatial and compositional microbial data (Castro et al., 2022b). Nevertheless, standardized protocols for optimized multiplex FISH in post-treatment BV analysis remain limited, particularly in longitudinal studies investigating recurrent disease. Therefore, further research is needed to refine FISH methodologies capable of accurately characterizing post-treatment vaginal microbial spatial organization. Addressing these methodological and knowledge gaps may improve understanding of BV persistence and facilitate the development of more targeted interventions aimed at restoring healthy vaginal microbial communities.

3. Aim

To characterize and compare spatial post-treatment vaginal microbial colonization patterns in antibiotic-responsive and antibiotic-resistant bacterial vaginosis using optimized multiplex fluorescence *in situ* hybridization (FISH).

4. Objectives

- To optimize and validate multi-species-specific FISH probes targeting *Lactobacillus* species and selected BV-associated anaerobes for the visualization and identification of vaginal microbial taxa.
- To quantify species-specific bacterial taxa pre- and post-treatment using PMA-qPCR.
- To correlate bacterial taxa composition and spatial organization with antibiotic response and resistance status in bacterial vaginosis post-treatment.

5. Materials and Method

5.1 Sample collection and preparation

This is the nested study within the recently completed CAPRISA 098 longitudinal cohort of women at risk for BV in Durban, KwaZulu-Natal (BREC/00006875/2024). All participants provided written informed consent and met the inclusion criteria, including BV positivity by Nugent score, absence of sexually transmitted infections, non-pregnant status, and no recent antibiotic use. Following enrollment, participants received metronidazole treatment and were followed for 24 days, with study visits conducted at 8-day intervals. For this study, samples will be stratified based on treatment outcomes into antibiotic-responsive (resolution to normal microbiota) and antibiotic-resistant BV (persistent BV post-treatment), enabling comparative analysis of spatial microbial colonization patterns.

5.2 Multiplex FISH Analysis

Multiplex FISH will be performed using probes targeting *Gardnerella* spp., *Lactobacillus iners*, *Lactobacillus crispatus*, other *Lactobacillus* spp., *Enterococcus* spp., *Atopobium (Fannyhessea)* spp., and a universal bacterial probe (Table 1). EChA fixative solution (ethanol:chloroform:acetic acid; 6:1:3) will be prepared and stored at 2-8 °C until use. Samples will be fixed by mixing with EChA solution at a 1:10 ratio and incubated overnight at 4 °C.

Microscope slides will be prepared by drawing two reaction chambers (~15 mm diameter) using a Pap pen and allowing them to air dry for at least 15 minutes. Slides will be placed on a flat surface, and 20 µL of fixed sample will be applied to each chamber and dried at 50 °C for 30 minutes. Hybridization will be performed by adding 50 µL of hybridization buffer (comprising nuclease-free water, 5 M NaCl, 1 M Tris-HCl pH 7.4, and 10% SDS) containing probes (0.5 µL of each probe at 50 ng/µL) to each chamber. Slides will then be incubated for 60 minutes at 50 °C in a humidified hybridization chamber.

Following hybridization, slides will be washed by briefly dipping 10 times in prewarmed (50 °C) sterile water, followed by incubation in prewarmed wash buffer (sterile water, 5 M NaCl, and 1 M Tris-HCl pH 7.4) for 5 minutes at 50 °C. Slides will be positioned back-to-back in 50 mL conical tubes during this step. After washing, slides will be air-dried under a fume hood in the dark.

For nuclear counterstaining, 200 µL of DAPI solution (1 µg/mL in 1× PBS) will be added to each reaction chamber and incubated for 10 minutes at room temperature in the dark. Slides will then be washed twice in deionized water by dipping 10 times per wash and subsequently air-dried. A drop of mounting medium will be added to each chamber, covered with a coverslip, and incubated overnight in the dark. Finally, slides will be visualized using a fluorescence microscope.

Table 1. FISH probes

Probe name	Sequence (5'–3')	Target organism	RNA target
Eub 338	GCT GCC TCC CGT AGG AGT	Most bacteria	16S rRNA
Gard 662	CCA CCG TTA CAC CGC GAA	<i>Gardnerella</i> spp	16S rRNA
Liner23-2	CTG CTC ACC TAS TTT CCG G	<i>Lactobacillus iners</i>	23S rRNA

Lab158	GGT ATT AGC A(T/C)C TGT TTC CA	<i>Lactobacillus</i> spp., <i>Enterococcus</i> spp.	16S rRNA
Lcrisp16-1	CGT CAT TAC CGA AGT AAA TCT GT	<i>Lactobacillus</i> <i>crispatus</i>	16S rRNA
Ato291	GGT CGG TCT CTC AAC CC	<i>Atopobium</i> spp	16S rRNA

Probe sequences adapted from Swidsinski et al., 2019

5.3 Propidium Monoazide (PMA)-qPCR for Differentiation between Live and Dead Cells

5.3.1 PMA treatment

Each sample (corresponding to those used for the FISH assay) will be processed for propidium monoazide (PMAxx)-based quantitative PCR (qPCR) to quantify viable bacteria. Samples will be aliquoted into two 1.5 mL microcentrifuge tubes (45 μ L each) and centrifuged at 8,000 rpm for 8 minutes, after which the supernatant will be discarded. The pellets will then be handled as follows: 50 μ L of 0.9% saline will be added to the control tube, which will be kept on ice. PMAxx working solutions will be prepared by diluting 20 mM PMAxx dye in 0.9% NaCl to obtain final concentrations of 50 μ M (Premix I) and 1 mM (Premix II).

For the first PMAxx treatment, 45 μ L of Premix I (50 μ M) will be added to the pelleted samples, followed by vortexing for 20 seconds (or thorough pipetting 10–20 times) to ensure proper mixing. Samples will then be incubated on ice for 15 minutes and subsequently exposed to light using a PMA-Lite LED Photolysis Device for 15 minutes. Immediately thereafter, a second PMAxx treatment will be performed by adding 2.25 μ L of Premix II (1 mM) directly to the samples. The tubes will again be vortexed for 20 seconds (or mixed by pipetting), incubated on ice for 15 minutes, and exposed to LED light for a further 15 minutes. A third PMAxx treatment will then be conducted by adding 2.25 μ L of Premix II, followed by the same sequence of vortexing, 15-minute incubation on ice, and 15-minute light exposure.

Following completion of the PMAxx treatments, samples will be centrifuged at 8,000 rpm for 8 minutes, the supernatant will be discarded, and the pellets will be resuspended in 45 μ L of 0.9% NaCl. Control samples maintained on ice will be centrifuged for 10 minutes at 8,000 rpm, and their supernatant will be removed. Finally, all samples will be stored at -20°C until genomic DNA extraction and subsequent qPCR analysis.

5.3.2 DNA Extraction from PMA experiment.

DNA extraction will be performed following three main steps: sample preparation, DNA extraction, and DNA quantification. Pellet samples obtained from the PMA experiment will be retrieved from storage and resuspended in 300 µL of sterile 0.9% saline solution. The samples will then be mixed thoroughly until complete homogenization of the pellets is achieved. DNA extraction will subsequently be performed using the MagAttract® PowerMicrobiome® DNA/RNA EP Kit according to the manufacturer's instructions. Samples will be loaded onto the Microlab STAR automated liquid handling system for automated extraction. The concentration of the extracted DNA will be measured using the Qubit® 3.0 Fluorometer following the manufacturer's protocol. The extracted DNA will then be stored at -20°C until further qPCR analysis.

5.3.3 Quantitative Polymerase Chain Reaction (qPCR)

Quantitative polymerase chain reaction will be utilized to amplify and quantify vaginal samples using QuantStudio™ 5 Real-Time PCR System (Thermo Fisher Scientific, USA Cat. no. A28574). Bacterial targets will be detected using fluorescence SYBR® Green dye designed specifically for each organism. 10 µL, which is the final volume, will be prepared for each reaction mixture, encompassing 0.5 µL, corresponding to the prime (forward and reverse primer) (Table 2), 5µL SYBR Green Master mix, 1 µL of the DNA extracted, and 3 µL of nuclease-free water. Quantification will be achieved through standard curves made through serial dilution of a known-concentration reference control. The thermal cycle program will be performed as follows: 95°C for 30 s, followed by 40 cycles of 95°C for 15 s, 60°C for 1 minute.

Table 2. Primer Sequences for PMA qPCR

Target Species	Forward Primer (5'–3')	Reverse Primer (5'–3')
<i>Gardnerella</i>	TCCCAACCCCAACTCACGATCTT	RCGCAAACCAACRATCTCAACTGG
<i>G. vaginalis</i>	TATTATAACTAAAGCTGCTG	TCGCCACTATAGTCG

<i>G. swidsinskii</i>	ATTTAGTTAGATATTTGGCAA	ATAGTCATATATTCCGCGC
<i>G. piovii</i>	AGCTGCTTACGATTATAGT	TTACTCATTCTAAGCTTAATAG
<i>G. leopoldii</i>	GATACTGCACTGTATCGA	CAGTATCAATACCAGCC
<i>Lactobacillus crispatus</i>	AGCGAGCGGAACTAACAGATTTAC	AGCTGATCATGCGATCTGCTT
<i>Lactobacillus iners</i>	GTCTGCCTTGAAGATCGG	ACAGTTGATAGGCATCATC

(Primers adapted from Tisakova et al., 2025)

6. Statistical Analysis

FISH images will be analyzed using ImageJ/Fiji and QuPath software to quantify bacterial abundance, fluorescence intensity, epithelial adherence, and spatial organization. Statistical analyses for PMA-qPCR data will be performed using GraphPad Prism version 11.0.1 (GraphPad Software, San Diego, CA, USA). Differences between antibiotic-responsive and antibiotic-resistant groups will be assessed using the Mann-Whitney U test. Correlations between viable bacterial burden and FISH-derived spatial organization parameters will be evaluated using Spearman's correlation analysis. Statistical significance will be set at $p < 0.05$.

7. Timeline



8. References

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